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Arba Minch University

COLLEGE OF SOCIAL SCIENCE AND HUMANITIES
DEPARTMENT OF GEOGRAPHY AND ENVIRONMENTAL

Geography of Ethiopia and The Horn



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UNIT ONE: INTRODUCTION

1.1. Geography: Definition, Scope and Themes

Meaning of Geography

- ✓ There is no single accepted definition by scholars b/c of the dynamic nature of the discipline and the changes in scope and method of study.
- ✓ The following may be accepted as a working definition.
- ✓ Geography is the *scientific study of the Earth that describes and analyses spatial and temporal variations of physical, biological and human phenomena, and their interrelationships and dynamism over the surface of the Earth.*

THE SCOPE, APPROACHES AND THEMES OF GEOGRAPHY

- ✓ **Geography is science** that explains the arrangements of various **natural** and **cultural** features on the Earth surface.
- ✓ Geography is a holistic and interdisciplinary field of study
- ✓ contributing to the understanding of the changing spatial structures from the past to the future.

THE SCOPE, APPROACHES AND THEMES OF GEOGRAPHY

- ✓ The scope of Geography is the surface of the Earth,
- ✓ which is the very thin zone that is the interface of the atmosphere, lithosphere, hydrosphere and biosphere,
- ✓ which provides the habitable zone in which humans are able to live.
- ✓ Geography can be approached by considering two continuums:
- ✓ a human-physical and a topical-regional continuum.

THE SCOPE, APPROACHES....

- ✓ The topical (systematic) Geography view categories of physical/human phenomena as distributed over the Earth
- ✓ **regional geography** is concerned with the associations within regions of all/some of the elements and their interrelationships.
- ✓ Geography has five themes namely **location**, **place**, **human-environment interaction**, **movement**, and **region**.

THEMES OF GEOGRAPHY

- ✓ **Location:** is particular place or position.
- ✓ Most studies of geography begin with the mention of this theme of geography.
- ✓ Location can be of two types: absolute location and relative location. In the former case, the location of a place is defined by its latitude and longitude or its exact address.

THEMES...

- ✓ **Place:** refers to the physical and human aspects of a location.
- ✓ It is associated with toponym (the name of a place),
- ✓ **site** (the description of the features of the place), and
- ✓ **situation** (the environmental conditions of the place).

THEMES..

- ✓ places in the world has unique characteristics
 - ✓ This is expressed in terms of **landforms**, **hydrology**, **biogeography**, **pedology**,
 - ✓ chxcs and size of its human population, and the distinct human cultures.
- ✓ The concept of “place” aids geographers to compare and contrast places on Earth.

HUMAN-ENVIRONMENT INTERACTION

- Humans have always been on ceaseless interaction with their natural environment.
- ✓ No other species that has lived on our planet has a profound effect on the environment as humans.
- ✓ Humans have adapted to the environment in ways that have allowed them to dominate all other species on Earth.
- ✓ human-environment interaction involves three distinct aspects:

HUMAN-ENVIRONMENT....

- ✓ **i) Dependency** refers to the ways in which humans are dependent on nature for a living.
- ✓ **ii) Adaptation** relates to how humans modify themselves, their lifestyles and their behavior to live in a new environment with new challenges.
- ✓ **iii) Modification** allowed humans to “conquer” the world for their comfortable living.

MOVEMENT

- It entails to the translocation of **human beings**, **their goods**, and their **ideas** from one end of the planet to another.
- ✓ The physical movement of people allowed the human race to inhabit all the continents and islands of the world.
- ✓ Another aspect of movement is the transport of goods from one place on the Earth to another.
- ✓ The **3Dimension** of movement is the flow of **ideas** that allows the unification of the human civilization and promotes its growth and prosperity.

REGION

- ✓ A region is a geographic area having distinctive chxcs that distinguishes itself from adjacent unit(s) of space.
- ✓ a formal region that is characterized by homogeneity in terms of a phenomenon (soil, temperature, rainfall, or other cultural elements like language, religion, and economy).
- ✓ a **functional or nodal** region chx.zed by functional inter/nships in a spatial system defined by the linkages binding particular phenomena.

1.2. LOCATION, SHAPE AND SIZE OF ETHIOPIA AND THE HORN

- ✓ The Horn of Africa is a narrow tip that protrudes into the northern Indian Ocean, separating it from the Gulf of Aden.
- ✓ It is the easternmost extension of African land
- ✓ the region that is home to the countries of Djibouti, Eritrea, Ethiopia, and Somalia,
- ✓ whose cultures have been linked throughout their long history.

1.2. LOCATION, SHAPE....

- ✓ Although the countries of the region share many common features,
- ✓ There is also great diversity among them, rendering each country unique in many respects.
- ✓ In terms of size, Ethiopia is the largest of all the Horn of African countries,
- ✓ while Djibouti is the smallest.

1.2.1. LOCATION OF ETHIOPIA

- ✓ The location of a country or a place on a map or a globe is expressed in two different ways.
- ✓ These are astronomical and relative locations (Figure 1.1).
- ✓ **Astronomical location**, also known as absolute or mathematical location, states location of places using the lines of latitudes and longitudes.

1.2.1. LOCATION...

- ✓ **Astronomically**, Ethiopia is a landlocked country
- ✓ located b/n 3°N (Moyale) and 15°N (Bademe - the northern most tip of Tigray) latitudes and
- ✓ 33°E (Akobo) to 48°E (the tip of Ogaden in the east) longitudes.
- ✓ The east west distance (15°) is longer than the north-south distance (12°).
- ✓ The latitudinal and longitudinal extensions are important in two ways.

1.2.1. LOCATION..

- ✓ First, as a result of its latitudinal extension the country experiences tropical climate and
- ✓ 2ndly due to its longitudinal extension there is a d/c of an hour b/n the most easterly and westerly points of the country.
- ✓ It is only for convenience that the +3 hours-time zone is used in all parts of the country.
- ✓ Relative location expresses the location of countries or places with reference to the location of other countries (vicinal), landmasses or water bodies.

1.2.1 LOCATION...

Table 1.1: The relative location of Ethiopia

Vicinal location	In relation to water bodies & land masses
▶ Sudan to the west and North west ▶ South Sudan to the south west ▶ Djibouti to the east ▶ Eritrea to the North and North east ▶ Kenya to the south	▶ In the horn of Africa ▶ South west of the Arabian Peninsula ▶ South of Europe ▶ Northwest of the Indian Ocean ▶ In the Nile Basin

1.2.1. LOCATION..

- ✓ The implications of the location of Ethiopia are described as follows.
- ✓ a) **Climate:** The fact that Ethiopia is located b/n 3°N and 15°N (b/n the Equator and Tropic of Cancer)
- ✓ implies that the country has a tropical climate, though modified by its altitude.
- ✓ The location of Ethiopia relative to the Indian Ocean, the Atlantic Ocean and the African and Asian landmass has also various bearings on the climate of Ethiopia.

1.2.1. LOCATION....

- ✓ **b) Socio-cultural:** Ethiopia is one of the earliest recipients of the major world religions namely Christianity, Islam and Judaism
- ✓ Its proximity to the Middle East, which was the origin of these religions.
- ✓ The linguistic and other cultural relationships, which Ethiopia shares with its neighbors, reflect the influence of location.

- ✓ c) **Political:** The political history of Ethiopia has been considerably influenced by: Geopolitical considerations of superpowers.
- ✓ Adjacency to the Red Sea (a major global trade route).
- ✓ The Middle East geopolitical paradigms.
- ✓ Ethiopia has been exposed for external invasions in a number of times; though the country resisted foreign intervention and remains free of external domination.

1.2.1. LOCATION..

Geography of Ethiopia and The Horn

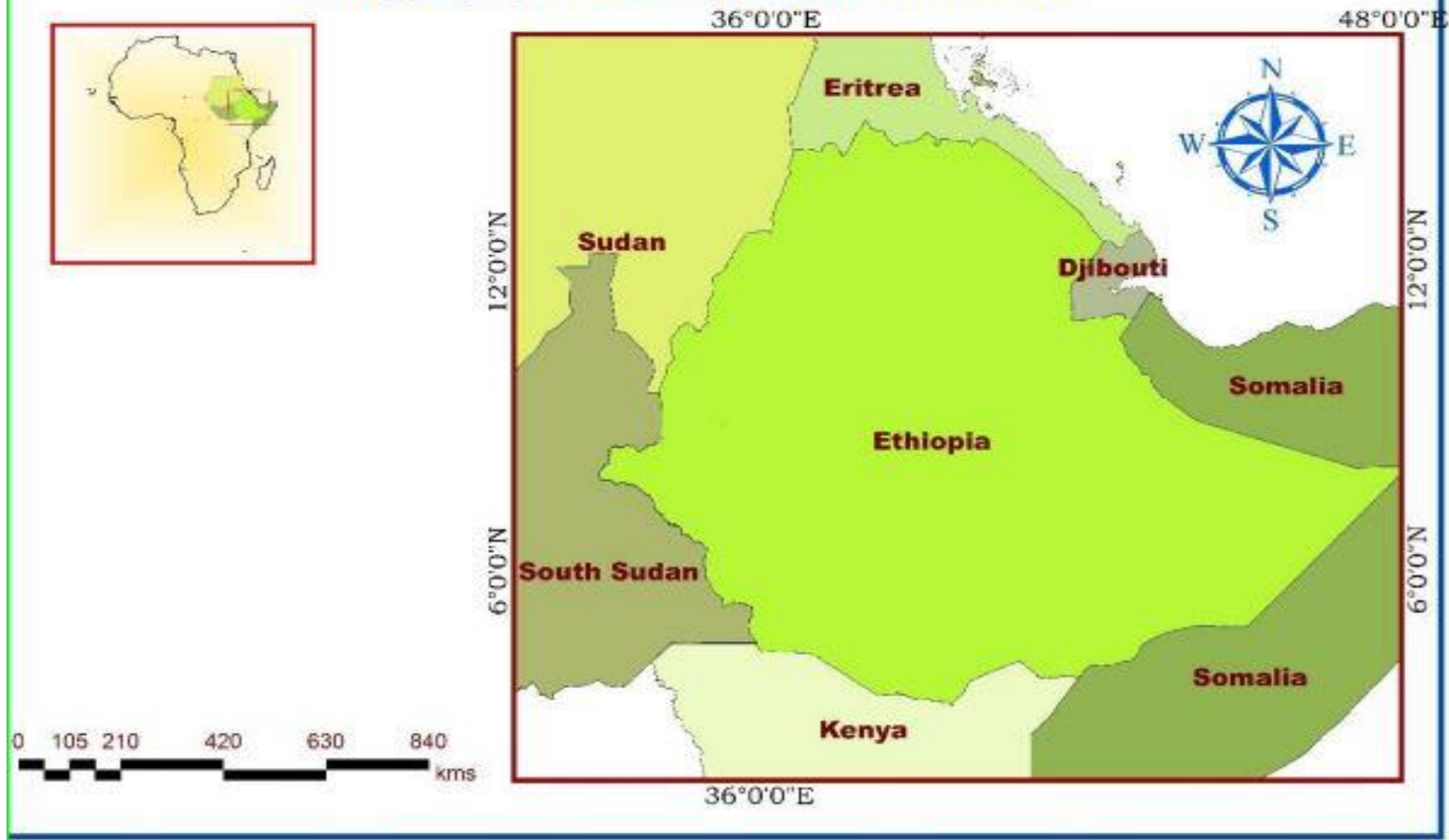


Figure 1.1. Ethiopia's location in relation to its neighboring countries

Source: Africa Map Library

1.2.2. SIZE OF ETHIOPIA

- ✓ Ethiopia with a total area of $\approx 1,106,000$ sqkm
- ✓ is the 8th largest country in Africa and
- ✓ 25th in the World.
- ✓ It extends about $\approx 1,639$ k.m East-West, and
- ✓ 1,577 k.m North-South.
- ✓ About 0.7% of the country is covered by water bodies.
- ✓ The size of Ethiopia also affects both the natural and human environment of the country.

1.2.2 SIZE..

Advantages

Disadvantage

- | | |
|--|---|
| ✓ Possess diverse agro ecological zones | ✓ Demands greater capital to construct infrastructural facilities |
| ✓ Variety of natural resources | ✓ Requires large army to protect its territory |
| ✓ Own extensive arable land | ✓ Difficult for effective administration |
| ✓ Have larger population size | ✓ Difficult for socio-economic integration |
| ✓ Home for diverse cultures | |
| ✓ Greater depth in defense extern invasion | |

THE SHAPE OF ETHIOPIA AND ITS IMPLICATION

- Countries of the World have d/t kinds of shape that can be divided into five main categories:
- ✓ compact, fragmented, elongated, perforated, and protruded.
- ✓ These shapes have implications on defense, administration and economic integration within a country.
- ✓ Whether some kind of shape is advantageous to a country or not, it depends on many other factors.

THE SHAPE

- ✓ **Compact shape countries:** The distance from the geographic Centre of the state to any of the borders does not vary greatly.
 - ✓ It is easier for defense, socioeconomic and cultural integration.
- ✓ **Fragmented shape countries:** They are divided from their other parts by either water, land/other countries.
- ✓ **Elongated shape countries:** - They are geographically long and relatively narrow like Chile.

THE SHAPE...

- ✓ **Perforated shape countries:** - A country that completely surrounds another country like the Republic of South Africa.
- ✓ There are d/t ways of measuring shape of countries.
- ✓ They are known as the indices of compactness.
- ✓ Indices measure the deviation of the shape of a country from a circular shape, which is the most compact shape.

THE SHAPE...

- ✓ No country with absolutely circular shape, nearly a circular shape are said to be more compact.
- There are four most commonly used measures of compactness. These are:
 - ✓ **1. Area-Boundary ratio (A/B):-** is ratio of an area of country to its boundary length.
 - ✓ The higher the A/B ratio, the greater the degree of compactness.

THE SHAPE...

- ✓ **2. Boundary-Circumference ratio:** the ratio of boundary length of a country to the circumference of circle having the same area as the country itself.
- ✓ It measures how far the boundary of a country verges on the circumference of a circle of its own size.
- ✓ Therefore, the nearer the ratio to 1 the more compact the country is

THE SHAPE...

- ✓ **3. Area-Circumference ratio (A/C):** - is ratio of an area of the country to the circumference of the smallest inscribing circle.
- ✓ It compares the area of the country with the circumference of a circle that passes touching the extreme points on the boundary of the country.
- ✓ The higher the A/C ratio, is the greater the degree of compactness.

THE SHAPE...

- ✓ **4. Area-Area (A/A') ratio:-** The ratio of the actual area of a country to the smallest possible inscribing circle.
- ✓ The area of the inscribing circle is the area of the smallest possible circle whose circumference passes through the extreme points on the boundary.
- ✓ Half-length of the longest distance between two extreme points gives radius of the inscribing circle.
- ✓ The nearer the ratio to 1, the more compact the country is.

THE SHAPE...

Table 1.3: Ethiopia's shape compared to its neighbors in the Horn.

Country	Area(km ²)	Boundary (km)	A/B ratio	B/C ratio	A/C ratio
Ethiopia	1,106,000	5,260	210.27	1.41	296.61
Djibouti	22,000	820	26.83	1.56	41.83
Eritrea	117,400	2,420	48.51	1.99	96.83
Kenya	582,644	3,600	161.85	1.33	215.28
Somalia	637,657	5,100	125.03	1.80	225.22

Source: Bekure (1982, 1983)

1.3 Basic Skills of Map Reading

- A map is a two-dimensional scaled representation of part or whole of the Earth surface on a flat body such as piece of paper, black board, wood or cloth.
- ✓ Map reading encompasses a systematic identification of natural features and manmade features.
- ✓ Natural features include mountains, plateaus, hills, valleys, river, ocean, rocks, plain etc.
- ✓ Man made features include roads, railway, buildings, dam etc.

BASIC SKILLS...

- ✓ Although many disciplines use maps, they have a special significance for Geographers as primary tools for displaying and analyzing spatial distributions, patterns and relations.
- ✓ Since these features cannot easily be observed and interpreted in real landscapes, maps are essential to Geographers.

IMPORTANCE OF MAPS

- ✓ Provide the basis for making geographical details of regions represented i.e. the geographical facts of an area such as relief, drainage, settlement etc.
- ✓ Maps are powerful tools for making spatial analysis of geographical facts of areas represented.
- ✓ Maps are useful for giving location of geographical features by varied methods of grid reference, place naming etc.

IMPORTANCE...

- ✓ Maps are used on disciplines like land use planning, military science, aviation, tourism, marine science, population studies, epidemiology, geology, economics, history, archaeology, agriculture etc.
- ✓ Map makes storage of the geographical data of areas represented.
- ✓ Maps are potentially used to assess' reliable measurements of the geographical features.
- ✓ The measurements can be of area size, distance etc

TYPES OF MAP

- There are many types of maps based on purpose and functions.
- ✓ Here topographical and statistical maps are considered.
- A. **Topographical maps:** depict one or more natural and cultural features of an area.
- ✓ Could be small, medium or large scale depending on the size of the area represented.

TYPES OF MAP

- ✓ Contents of topographical maps depend on purpose of a map, scale of a map, date of compilation, and nature of the land represented.

B. Special purpose/statistical maps: These are maps, which show distribution of different aspects such as temperature, rainfall, settlement, vegetation etc.

MARGINAL INFORMATION ON MAPS (ELEMENTS OF MAPS)

- ✓ It is shown on a map to enable the reading and interpretation of the geographical information of an area represented. This includes:
 - ✓ **a. Title:** It is the heading of the given map which tells what the map is all about.
 - ✓ **b. Key (legend):** It is the list of all conventional symbols and signs shown on the map with their interpretation.

MARGINAL INFORMATION...

- c. **Scale:** It is the ratio b/n the distance on the map and the actual ground distance.
- ✓ Scales enable the map user to interpret the ground measurement like road distance, areal sizes, gradient etc.
- ✓ It can be expressed as representative fraction, statements/verbal scale, and linear (graphic) scale.

d

MARGINAL...

d. North arrow: It is indicated with the north direction on a map; used to know the other important directions of the mapped area like east, west, south, and west.

e. Margin: Is the frame of the map.

✓ It is important for showing the end of the mapped area.

f. Date of compilation: It is a date of map publication.

✓ Enables users to realize whether the map is updated/outdated.

BASIC PRINCIPLES OF MAP READING

- basic principles of map reading abilities and skills applied by map-reader to translate map symbols into landscape images.
- ✓ Map Readers must have ideas about the symbol and the real World (landscapes).
- ✓ Every map symbol visualized by the reader to read a map.

BASIC SKILLS...

- ✓ The first symbols introduced should be those, which refer to landscape features of which the reader already has some images.
- ✓ Map symbols are introduced as needed.
- ✓ 2ndly, knowledge of directions is an important principle in reading maps.

BASIC SKILLS...

- ✓ A basic functions of maps is to orient ourselves and locate places on the earth.
- ✓ Unless a reader knows the basic directions, he or she may not use a map effectively.
- ✓ Before locating features using a map, north (the north arrow) should be determined and the readers should have practice in finding this direction.

BASIC SKILLS...

- ✓ learning to read direction on a map and the location of the features on the maps in r/n to one another.
- ✓ Maps are covered in a series of lines that make up a grid
- ✓ The lines have numbers allow you to accurately pinpoint your location on a map.
- ✓ the grid system makes simple to give an accurate description of your location.
- ✓ This description, which will be a series of numbers, is known as a grid reference.

CHAPTER TWO

THE GEOLOGY OF ETHIOPIA AND THE HORN

2.1 Introduction

- **Geology** is science of Earth that studies the evolution, the materials composition, and the processes acting upon Earth.
- ✓ It is concerned with events that took place in the remote past when no witness and with features far beneath the earth's surface where no one can see.
- ✓ A great deal of geological understanding obtained by inference from what can be seen and what can be measured.

INTRO.....

- ✓ Many clues not only the rocks and landforms which can be observed and studied at the Earth's surface,
 - ✓ but also those provided by indirect methods such as **geophysics** (e.g. studying earthquake waves which can penetrate deep beneath the earth's surface),
 - ✓ **geochemistry** (analysis of the detailed composition of rocks which can give clues as to their origin) and
 - ✓ **geochronology** (methods for finding the ages of rocks, from the radioactive elements they contain).

INTRO...

- The earth's continents were together in a single huge continent called **Pangaea**.
- ✓ The large super continent was then split into **Gondwanaland** where Africa is a part and **Laurasia**;
- ✓ later into smaller fragments over the last million years.
- ✓ These then drifted apart to form the present arrangement of continents.

INTRO.....

- ✓ German Climatologist Alfred Wegener proposed the hypothesis that the continents were assembled together as a supercontinent, called **the Continental drift Theory**.
- ❖ **Wegener's principal observations were:**
 - ✓ **Fit of the continents:** The opposing coastlines of continents often fit together.
 - ✓ **Match of mountain belts, rock types:** If the continents are reassembled as Pangaea, mountains in West Africa, North America, Greenland, and Western Europe match up.

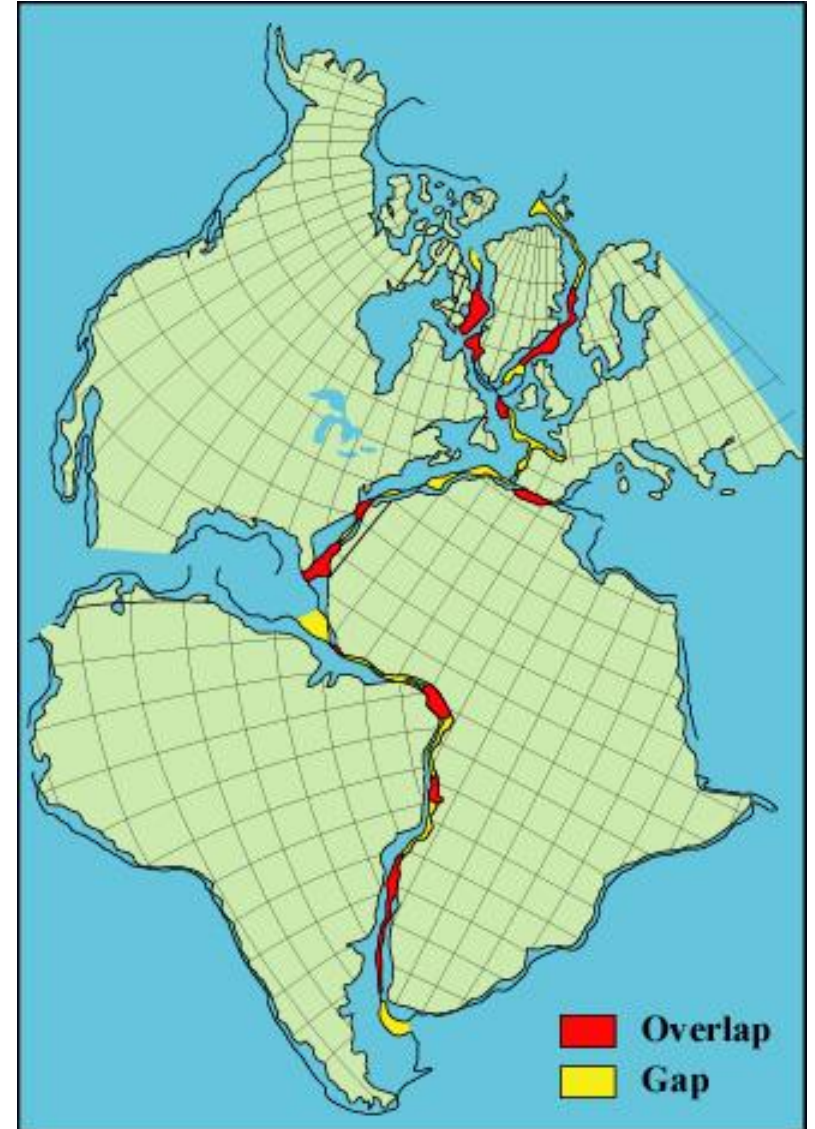
INTRO....

- ✓ **Distribution of fossils:** The distribution of plants and animal fossils on separate continents forms definite linked patterns if continents reassembled.
- ✓ **Paleoclimates:** rocks formed 200 million years ago in India, Australia, South America, and southern Africa all exhibited evidence of continental glaciations.

INTRO....



- fit of continental margins is best when the outline is the edges of the continental shelves.



2.2. THE GEOLOGIC PROCESSES: ENDOGENIC AND EXOGENIC FORCES

- Geology studies of how Earth's materials, structures, processes and organisms have changed over time.
- ✓ These processes are divided into two major groups: internal and external processes.
- ✓ The internal processes (endogenic) include volcanic activity and all the tectonic processes (folding, faulting, orogenesis (mountain building), and epeirogenesis (slow rising and sinking of the landmass)).

THE GEOLOGIC PROCESSES.....

- ✓ The processes result in building of structural and volcanic features like plateaus, rift valleys, Block Mountains, volcanic mountains, etc.
- ✓ The external (exogenic) processes are geomorphic processes.
- ✓ They include weathering, mass transfer, erosion and deposition.

THE GEOLOGIC PROCESSES....

- ✓ They act upon the volcanic and structural landforms by modifying, roughening and lowering them down.
- ✓ The landmass of Ethiopia, as elsewhere, is the result of the combined effect of endogenic and exogenic processes.

DATING TECHNIQUES

- The geological history is divided into Eras.
- ✓ Each Era is divided into periods.
- ✓ The Eras are given names that indicate the kind of **life** that existed in them. For instance,
 - the **Paleozoic Era (ancient life)** is the age of invertebrates,
 - the **Mesozoic Era (the middle life)** is the age of reptiles
 - while the **Cenozoic Era (recent life)** is the age of mammals.

2.3. GEOLOGICAL...

- ✓ These geological time divisions differ from each other in such characteristics as
 - the relative position of land and sea,
 - kind of climate and important animal and
 - plant life that developed and existed at that Era/period.
- ✓ Geological time is difficult to measure precisely.

2.3. GEOLOGICAL...

- The Earth was formed ≈ 4.5 billion years ago and the earliest forms of life were thought to have originated about 3.5 billion years ago.
- ✓ The first major boundary is defined by what was believed at the time to be the first appearance of life on earth.
- ✓ The **span of time before life** appeared was termed the **Precambrian Era**.

2.3. GEOLOGICAL...

- ✓ To describe the geology and history of life on Earth, scientists have developed the geological time scale.
- ✓ The geological time scale measures time on a scale involving four main units:
 1. An epoch is the smallest unit of time on the scale and encompasses a period of millions of years.

2.3. GEOLOGICAL...

2. Chronologically, epochs are clumped together into larger units called periods.
3. Periods are combined to make subdivisions called Eras.
4. An eon is the largest period of geological time.

2.3. GEOLOGICAL...

- ✓ The division of time units in the geological time scale is based on the occurrence geological events (e.g. mass extinctions).
- ✓ the geological time categories do not consist of a uniform length of time.
- ✓ The geological time scale is built largely on the basis of life and evolution

TABLE 2.1: THE GEOLOGICAL TIME SCALE

Era	Period	Began (million years)	Ends	Major events(Million years ago)
4. Cenozoic	Quaternary	1.6	present	Major glaciers in North America and Europe (1.5)
	Tertiary	70	1.6	Rocky Mountains (65), individual continents take shape.
3. Mesozoic	Cretaceous	146	70	Dinosaurs extinct (65), western interior seaway and marine reptiles (144 – 65)
	Jurassic	208	146	Pangaea (one land mass) begins to

2.3. GEOLOGICAL...

2. Paleozoic	permian	290	225	Greatest extinction on Earth (245)
	Pennsylvanian	322	290	Greatest extinction on Earth (245)
	Mississippian	362	322	Coal-forming forests
	Devonian	408	362	First land animals and first forests (408)
	Silurian	439	408	Life invades land
	Ordovician	510	439	First fish appeared
	Cambria	600	510	Great diversity of marine invertebrates
1. Precambrian	Proterozoic	2,500	600	Marine fossil invertebrates (600)
	Archean	4,500	2,500	Earliest fossils recorded (3,500), earliest rock formation (4,000)

AGE DATING TECHNIQUES

- There are two techniques of knowing the age of rocks:

A. Relative Dating

- ✓ It uses geological evidence to assign comparative ages of fossils.
- ✓ we can use two ways to know the relative age of a rock:
 - i) By look at any fossils the rock may contain. If any of the fossils are unique to one of the geologic time periods, then the rock was formed during that particular time period.

AGE DATING ...

- ii) using "What is on top of the older rocks?"
- ✓ layers of rocks in a cliff or hillside, younger rocks are on top of older rocks.
- ✓ these methods only give the relative age of rocks -which one is younger and which is older.

B. Absolute Dating: known as Radiometric techniques.

- ✓ Technique developed with discovery of radioactivity in 1896.

ABSOLUTE...

- ✓ The regular rates of decay for unstable, radioactive elements were found to constitute virtual “clocks” within the earth’s rocks.
- ✓ Radioactive elements like uranium (U) & thorium (Th) decay to form different elements/isotopes of the same element.
- ✓ Every radioactive element has its own half-life.

B. ABSOLUTE DATING...

- At the end of the period constituting one half-life, half of the original quantity of radioactive element has decayed; after another half-life, half of what was left is halved again, leaving 1/4th of the original, etc...

Two of the major techniques include:

B. ABSOLUTE DATING...

- ✓ ii. **Carbon-14 Technique:** Upon the organism's death, carbon-14 begins to disintegrate at a known rate, and no further replacement of carbon from atmospheric carbon dioxide can take place.
- ✓ Carbon-14 has half-life of 5730 years.
- ✓ ii. **Potassium-Argon Technique:** is widely used for dating rocks.

B. ABSOLUTE DATING...

- ✓ Geologists are able to date entire rock samples b/c potassium-40 is abundant in micas, feldspars, and hornblendes.
- ✓ Leakage of argon is a problem if the rock has been exposed to temperatures $>125^{\circ}\text{C}$ (257°F),
- ✓ b/c the age of the rock will then reflect the last episode of heating rather than the time of original rock formation.

LANDFORMS OF ETHIOPIA AND THE HORN

2.4.1. The Precambrian Era Geologic Processes (4.5 billion - 600 million yrs).

- It covers 5/6th of the earth's history.
- ✓ remoteness in time and the absence of well-preserved fossils, our knowledge of the events is limited.
- ✓ general description of main geologic processes can be made.
- ✓ The major geologic event of the Precambrian Era was Orogenesis.

2.4.1 PRECAMBRIAN

- ✓ As a result, the land was subjected to intense folding.
- ✓ This was accompanied by intrusive igneous activity.
- ✓ The result was the formation of huge mountain ranges
- ✓ In b /n the orogenic periods and after the last orogenesis, there were long periods of denudation, which reduced these mountains to near-level (peneplained) rock surfaces.

2.4.1 PRECAMBRIAN

- ✓ This “levelled” surface was later (in the Mesozoic and Cenozoic Eras) covered by younger rock formations
- ✓ Therefore, in most parts of Ethiopia rocks belong to this Eras found beneath all others, forming the basement rocks.

2.4. 1 THE PRECAMBRIAN...

- ✓ they had been subjected to pressure and heat from overlying weight, earth movements (folding, orogenesis) and to intrusive igneous activity;
- ✓ the original rocks (both sedimentary and igneous) were altered into metamorphic rocks of varying stages of metamorphism.

2.4.1 PRECAMBRIAN

- ✓ Since these same processes have allowed mineralization and crystal formation,
- ✓ They are collectively described as crystalline rocks.
- ✓ The Precambrian rocks are overlaid by recent rock formations
- ✓ surface rocks covering 25% of the land mass of the country;

2.4.1 PRECAMBRIAN

- ✓ They found exposed in the northern part: Western lowlands, parts of northern and central Tigray.
 - A. In the western Part: Gambella, Benishangul-Gumuz (Metekel and Asossa), western Gojjam, western Wellega, Illuababora, and Abay gorge.
 - B. In the southern Part: Guji, southern Omo, and parts of southern Bale and Borena.
 - C. In the eastern part: Eastern Hararghe.

MILLION - 225 MILLION YEARS AGO)

- It lasted for about 375 million years.
- ✓ The major geological process was denudation.
- ✓ The gigantic mountains that were formed by this Era orogeny were subjected to intense and prolonged denudation.
- ✓ Later, the gigantic mountain ranges were reduced to a “peneplained” surface.

2.4.2 THE PALAZOIC...

- ✓ Undulating plain of residual features (inselbergs) here and there was formed.
- ✓ The sediments were transported southward and eastward to form continental (in Africa) and marine deposits, respectively.
- ✓ Because of the limited deposition within Ethiopia, rocks belonging to this Era are rare in the country

2.4.3. The Mesozoic Era Geologic Processes (225-70 million years ago)

- This Era lasted for about 155 million years.
- It was an Era of alternate slows sinking and rising (epeirogenesis) of the landmass.
- This process affected the whole present-day Horn of Africa and Arabian landmass.
- At the same time the land was tilted eastward and lowered in the southeast and higher in the northwest.

2.4.3 THE MESOZOIC...

- ✓ subsidence of the land began at 225 million years ago.
- ✓ As the land sank slowly the sea invaded, from Somalia and Ogaden and slowly spreading northwestward.
- ✓ This was in late Triassic period.
- ✓ This phenomenon continued up to Jurassic period.
- ✓ As the shallow sea spread towards the land, sands were deposited over the peneplained Precambrian rock surface.

2.4.3 THE MESOZOIC...

- ✓ By means of the depth of the sea increased, mud (shale), gypsum and later lime were deposited.
- ✓ The latter is associated with the flourishing of marine life and decaying and precipitating of their remains, as the sea stayed long.
- ✓ Hence, Mesozoic rocks are considered to have the greatest potential for oil and gas deposits.

2.4.3 THE MESOZOIC...

- Through time, compression by the overlying rocks and by cementing minerals, the sands and lime were compacted to form sandstone and limestone layers respectively.
- ✓ These are known as the Adigrat sand stone and Hintalo limestone layers.
- ✓ They are named after place names in Tigray where they might have been first identified.

2.4.3 THE MESOZOIC...

- In the Horn of Africa and Ethiopia, the slow rise of the land and consequently the regression of the sea began in the Upper Jurassic.
- ✓ It continued throughout the Cretaceous period.
- ✓ With the retreat of the sea, another process of deposition occurred.

2.4.3 THE MESOZOIC...

- ✓ In the country sedimentation ended with the deposition of clay, silt, sand conglomerate brought in from the land as the sea receded due to uplift of the landmass.
- ✓ Gypsum, shale and at last sands were laid over the Hintalo limestone.
- ✓ The uppermost layer is known as the Upper sandstone

2.4.3 THE MESOZOIC....

- ✓ By the end of the Mesozoic Era, when the land emerged out of the sea, three major sedimentary formations were laid and formed upon the Precambrian rock surface.
- ✓ These were the Adigrat or lower sandstone, Hintalo limestone and Upper Sandstone

2.4.3 THE MESOZOIC...

- ✓ As transitional formations, gypsum and shale were inter-bedded above and below the Hintalo limestone.
- ✓ The Mesozoic sedimentary rocks cover 25% of the land mass of the country.

2.4.3 THE MESOZOIC ...

- ✓ Tilting of the landmass during the transgression and regression of the sea, and direction of the invading and retreating sea, the age and thickness of the Sandstone layers vary in a Southeast - Northwest direction.
- ✓ The Adigrat sandstone is older and thicker in the southeast and progressively decreases in age and thickness northwestward.

2.4.3 THE MESOZOIC ...

- ✓ The Upper sandstone is thicker and younger (Upper Cretaceous) in the Southeast, while in the Northwest it is older and thinner
- ✓ The transgressing sea and Mesozoic sediments nearly covered the whole of Ethiopia.
- ✓ The northwestern limit was as far as central Tigray, and western slopes of Western highlands.

2.4.3 THE MESOZOIC...

- ✓ In most parts of Ethiopia, the Mesozoic rocks are overlaid by the Cenozoic rocks.
- ✓ As surface rocks, these old marine sediments are extensively found in the Southeast lowlands.
- ✓ Other exposures include central Tigray, and along the gorges of Abay and Wabishebele rivers.

2.4.3 THE MESOZOIC...

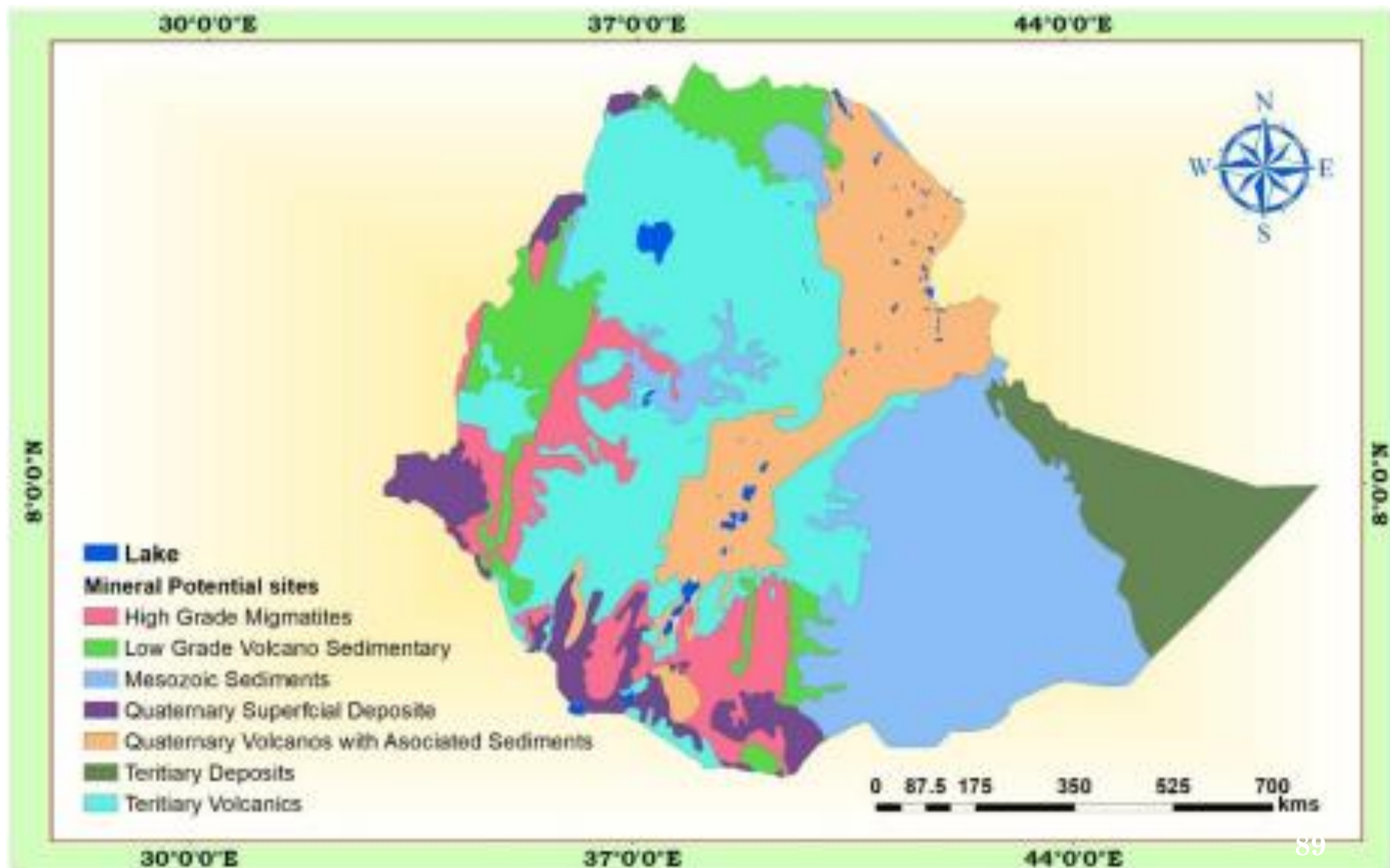


Figure 2. 1. Geological map of Ethiopia Source: CSA Shapefile, 2012

2.4.4. THE CENOZOIC ERA GEOLOGIC PROCESSES (70MILLION YEARS AGO - PRESENT)

- It is the most recent of the geologic Eras.
- ✓ The tectonic and volcanic activities of this Era have an important effect in the making of the present-day landmass of Ethiopia and the Horn of Africa.
- ✓ The land was subjected to two major geologic events and other geologic processes of lesser magnitude but still important.

2.4.4 THE CENOZOIC...

- ✓ These geologic activities are:
 - a. Uplifting of the Arabo-Ethiopian landmass and outpouring of huge quantity of lava.
 - b. Formation of the Rift Valley.
 - c. Quaternary volcanism and deposition
- ❖ **Uplifting of the Arabo-Ethiopian landmass and outpouring of lava flood**
- ✓ The uplifting of the whole of the Arabo-Ethiopian landmass is a continuation of the slow rise that began in the Upper Jurassic and Cretaceous periods.

2.4.4 THE CENOZOIC....

- This huge uplift continued to the Paleocene and Oligocene epoch of the Tertiary period.
- ✓ Where the uplifting was of greater magnitude, the land was pushed up to a maximum height of 2,000 meters above sea level.
- ✓ This occurred during the Eocene epoch.
- ✓ This uplifting was of an epeirogenic character.
- ✓ It was unparalleled anywhere else except in the Alpine orogenic belts of the Andes

2.4.4 THE CENOZOIC...

- ✓ The whole of the Arabo-Ethiopian landmass was pushed up in blocks as one mass.
- ✓ The greatest uplift was in central Ethiopia.
- ✓ This immense tectonic force also fractured the crust at many places.
- ✓ Huge quantity of lava came out through these fractures.

2.4.4 THE CENOZOIC...

- ✓ The out pouring of this flood of basalt spread widely and extensively and covered a large part of the Mesozoic sedimentary layer to form the Ethiopian plateau surface and the floor of the present-day Rift Valley.
- ✓ At that time the Rift Valley was not yet formed
- ✓ The mass of lava was so immense, that it formed a thick layer of volcanic rocks on the plateau, which is uplifted to more than 1,000m.amsl in the north Central Highlands.

2.4.4 THE CENOZOIC...

- ✓ Where the lava comes out through vents, huge volcanic rocks were piled up to form many and huge Volcanic Mountains' towering the flat basaltic plateau.
- ✓ This volcanic material is known as Trappean lava or Trap Series lava.
- ✓ Where it is not dissected by erosion, the Ethiopian plateau had flat and nearly horizontal surface.
- ✓ This is an expression of the peneplaned Precambrian surfaces, the smothering effect of the Mesozoic deposition, and the spreading of the Trap series lava over the Mesozoic sediments.

2.4.4 THE CENOZOIC...

- ❖ The Formation of the Rift Valley
- ✓ The formation of the Rift Valley is said to be related with the theory of plate tectonics.
- ✓ Theoretically, the Rift Valley lying on the Earth's crust below which lateral movement of the crust in opposite directions producing tensional forces that caused parallel fractures or faults on the sides of the up-arched swell.

2.4.4 THE CENOZOIC...

✓ As the tension widened the fractures, the central part of the landmass collapsed to form an extensive structural depression known as the

Rift Valley (Fig. 2.2).

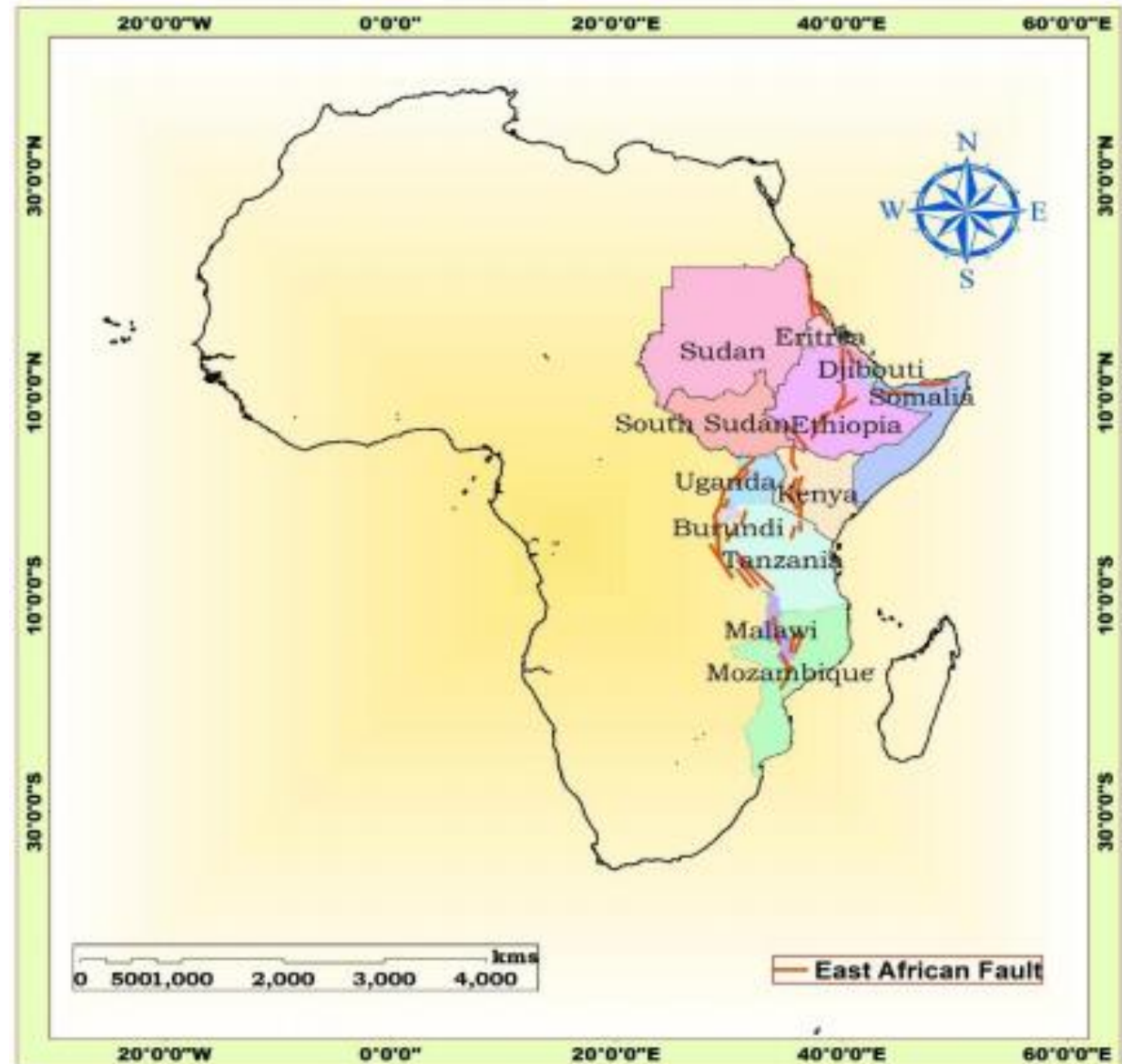


Figure 2.2. The Ethiopian and East African Rift Valley systems.
Source: Africa Map Library

2.4.4 THE CENOZOIC...

- The major faulting movement probably began in the late Oligocene and Miocene Epochs of the Cenozoic Era.
- ✓ This rifted the Red Sea trough, which began to be flooded from the north.
- ✓ But the major rifting, affecting the whole African Rift System, including that of Ethiopia and the Gulf of Aden took place in the Miocene Epoch.
- ✓ Rifting and faulting, however, continued all the time throughout the Pliocene and even the Pleistocene Epochs.

2.4.4 THE CENOZOIC...

- The Red Sea and the Gulf of Aden were connected as a result of the rifting and faulting of the land bridge that separated them.
- ✓ At the same period (Pliocene), the Afar depression (with Gulf of Zula) was down-faulted allowing the Red Sea water to penetrate far inside.
- ✓ Reversed tilting and volcanic activity, later (Pleistocene) blocked the connection and isolated the extension of the sea, allowing much of the water to evaporate.

2.4.4 THE CENOZOIC...

- ✓ As a result, thick saline materials accumulated.
- ✓ In the same period, the area be/n the Danakil Depression and the Red Sea was uplifted to form the Afar Block Mountains.

The Spatial Extent of the Rift Valley

- It is part of the Great East African Rift system that extends from Palestine-Jordan in the north to Malawi-Mozambique in the south,
- ✓ Cover distance of about 7,200 kilometers.
- ✓ 5,600 kilometers is in Africa, and
- ✓ 1,700 kilometers in Eritrea and Ethiopia.
- ✓ the widest part of the Rift Valley is the Afar Triangle (200-300 km).

THE SPATIAL EXTENT...

- ✓ The Red Sea, the Gulf of Aden, and the East African System meet and form the triangular depression of the Afar where the Kobar Sink lies about 125 meters below sea level.
- ✓ The formation of the Gulf of Aden and the separation of the Arabian Peninsula from the Horn of Africa also took place during the Tertiary period.

THE SPATIAL EXTENT...

- ✓ The Rift Valley region of Ethiopian is the most unstable part of the country.
- ✓ There are numerous hot springs, fumorales, active volcanoes, geysers, and frequent earthquakes.
- ✓ The formation of the Rift Valley has the following structural (physiographic) effects:

THE SPATIAL EXTENT...

- ✓ It divides the Ethiopian Plateau into two.
- ✓ It separates the Arabian landmass from African landmass.
- ✓ It causes the formation of the Dead Sea, Red Sea and the Gulf of Aden troughs.
- ✓ It creates basins and fault depressions on which the Rift Valley lakes are formed.

THE SPATIAL EXTENT...

- ✓ Faulting and graben formation are not only limited to the Rift Valley.
- ✓ similar tectonics activities have occurred in the Lake Tana Basin.
- ✓ However, the formation of Lake Tana had been accentuated by volcanic activity so that lava flow in the southeast had dammed part of the rim to deepen the basin.
- ✓ Faulting in other places had a structural control along some part of the river courses

QUATERNARY VOLCANIC ERUPTIONS AND DEPOSITIONS

- ✓ They are recent volcanic activities that took place after the formation of the Rift Valley.
- ✓ This occurred in the Pliocene-Pleistocene Epochs.
- ✓ This is a continuation of the tectonic and volcanic processes that earlier affected the Rift Valley formation.
- ✓ It occurred in the form of renewed rifting/faulting and more volcanism.

QUATERNARY....

- ✓ This activity was generally limited to the floor of the Rift Valley and the region south of Lake Tana, where the lava covers an area of more than 3,000km² Aden volcanics and
- ✓ recent faulting are more extensively developed in the Afar region.
- ✓ The area is shattered by numerous faults and subjected to extensive scoriaceous basalt eruptions.

QUATERNARY....

- ✓ The latter phenomenon is also widely manifested in the main Ethiopian Rift, especially in its northern section.
- ✓ Some of the eruptions have occurred in the last few hundred years.
- ✓ B/c of their recent occurrence, the Aden volcanics have relatively well-preserved and visible morphological features.

QUATERNARY....

- ✓ The basic volcanic features of the Aden series include the following:
 - ✓ Numerous and freshly preserved volcanic cones, many of which have explosive craters.
 - ✓ Some of these are active Dubi, Erta Ale, Afrera etc. Of these, Erta Ale is the most active volcano in Ethiopia.

QUATERNARY....

- ✓ Volcanic hills and mountains, some of which are semi-dormant (Fantale, Boseti-Gouda near Adama, Aletu north of Lake Ziway, Chebbi north of Lake Hawassa etc.).
- ✓ Extensive lava fields and lava sheets some of which are very recent.
- ✓ Lava ridges.
- ✓ Thermal springs, fumaroles etc.

QUATERNARY DEPOSITION

- During the Quaternary period of the Cenozoic Era, the Earth experienced a marked climatic change, where warmer and dry periods were alternating with cooler and wet periods.
- This was the time of the last “Ice Age” in the middle and high latitude areas and the time of the “Pluvial Rains” in Africa.
- The heavy Pluvial Rains eroded the Ethiopian plateau and the eroded materials were deposited in the Rift Valley lakes.

QUATERNARY DEPOSITION...

- ✓ The excessive rain resulted in an excessive surface flow; rivers were many and large.
- ✓ They carried a lot of water and sediments. Lake and marshy areas became numerous and deep.
- ✓ Many were enlarged and covered much area and even merged together.
- ✓ For example, Ziway-Langano- Shalla; Hawasa-Shallo; Chamo-Abaya; and Lake Abe and the nearby smaller lakes and marsh basins formed huge lakes.
- ✓ After the ‘‘Pluvial Rains’’, the Earth’s climate became warmer and drier.

QUATERNARY DEPOSITION...

- ✓ Thus, it increased the rate of evaporation that diminished the sizes of the lakes.
- ✓ Today, there are lacustrine deposits of continental origin around many of the Ethiopian lakes, river valleys and lowlands.
- ✓ According to the place and manner of deposition and depositing agents these deposits are divided as follow.
 - a. Lacustrine deposits: Deposits on former lakebeds, and swampy depressions

QUATERNARY DEPOSITION...

- b. Fluvial deposits: Deposits on the banks of rivers, flood plains both in plateau, foothills etc
- ✓ c. Glacio-fluvial deposits and erosional features: These are occurred on high mountains, such as Bale and Kaka Mountains.
- ✓ d. Aeolian deposits: Are windblown deposits.
- ✓ e. Coastal and marine deposits: Deposits on sea invaded and sea-covered places.

QUATERNARY DEPOSITION...

- ✓ The quaternary deposits are mainly found in the Rift Valley (Afar and Lakes Region), Baro lowlands, southern Borena, and parts of northwestern low lands.
- ✓ Cenozoic rocks cover 50% of the land mass of the country.
- ✓ These include Highland Tertiary volcanics (basalts), Tertiary as well as Quaternary volcanics, and sediments of the rift valley.

2.5. ROCK AND MINERAL RESOURCES OF ETHIOPIA

- ✓ The occurrence of metallic minerals in Ethiopia is associated with the Precambrian rocks.
- ✓ Although not in sufficient concentration and extent, a great variety of such minerals occur in the basement rocks. These rocks contain most of the metallic deposits known at present.
- ✓ The exploitation and search for mineral deposits in Ethiopia has been taking place for the past 2,000 years or so, and its early cultures were based partially on the mineral wealth of the day.

ROCK AND MINERAL....

- ✓ Such has been the case of gold production and utilization, which has become part of Ethiopia's history, tradition and folklore.
- ✓ The mining and working of iron for the manufacture of tools, utensils and weapons, and the use of salt and salt-bar all these indicate to a long mining tradition.
- ✓ presently mineral production from Ethiopia has been negligible by World standards.

2.5.1. BRIEF FACTS AND CURRENT STATE OF MAIN MINERALS IN ETHIOPIA

- ✓ Geological surveys proved that Ethiopia has abundant mineral resources of metals and precious metals, coal, and industrial minerals.

Gold

- ✓ Gold has been mined in Ethiopia for quite long time, mainly from Benishangul-Gumuz (Metekel) and Adola.
- ✓ Operating mines produce gold from primary sources in such localities as Dermidama, Sakoro and Lega-dembi.

GOLD...

- ✓ Mechanised alluvial working is confined to the state- operated gold field of Adola.
- ✓ 2ndary gold deposits are common in the following localities:
 - ✓ Adola,
 - ✓ Murmur Basin,
 - ✓ Shakiso,
 - ✓ Awata Basin,
 - ✓ Dawa Basin,
 - ✓ Ghenale Basin,
 - ✓ Ujama Basin,
 - ✓ Makanisa (Guba and Wombera),
 - ✓ Kaffa.
 - ✓ In Gambella and Illuababora (Akobo River),
 - ✓ in Sidama (Wondo),
 - ✓ Borena (Negele-Yabelo area) and
 - ✓ in Benishangul-Gumuz (Sherkole),
 - ✓ west Wellega,
 - ✓ Mengi-Tumat-Shangul areas to the Sudanese border, and the drainage of the Didessa and Birbir

PLATINUM

- ✓ The Yubdo area in Wellega, is the only active Ethiopian Platinum mine.
- ✓ Platinum occurrences have been reported from Delatti in Wellega, and the valley of Demi-Denissa and Bone Rivers as well as Tullu Mountain area in Sidama.

TANTALUM

- Significant deposit of tantalum and niobium is found in southern Ethiopia.
- ✓ It occurs in Adola area where Kenticha Tantalum mine with resources of more than 17,000 metric tons of world class ore reserve is found.
- ✓ The sedimentary and volcanic rock activities are also resourceful.

TANTALUM

- ✓ Extensive lignite deposits in Ethiopia are found in Nedjo (Wellega), and in small amounts in Chilga (Gonder) are found in the sedimentary formations laid in b/n Trapean lava.
- ✓ important Lignite, one of the lowest ranked coal, is known to occur in many localities such as
- ✓ in the Beressa Valley and Ankober (North Shewa),
- ✓ Sululta (near Addis Ababa),

TANTALUM...

- ✓ Muger Valley (West Shewa),
- ✓ Aletu valley (near Nedjo),
- ✓ Kariso and Selmi Valleys (Debrelibanos),
- ✓ Zega wodem gorge (near Fiche), Didessa Valley (southwest of Nekemte),
- ✓ Kindo and Challe Valley (Omo confluence),

TANTALUM...

- ✓ Adola, Wuchalle (north of Dessie),
- ✓ Chukga area (on Gonder-Metema road),
- ✓ Dessie area (near Borkena River).
- ✓ These areas are promise to be a good prospect to meet some of the local industrial and domestic needs.

GEMSTONES

- It includes amethyst,
 - aquamarine,
 - emerald,
 - garnet,
 - opal,
 - peridot,
 - sapphire, and tourmaline
- ✓ occur in many parts of Ethiopia, mainly in Amhara and Oromia Regional States.

GEMSTONES ...

- ✓ Quality Opal was first discovered by local people in Wadla and Dalanta woredas, North Wello in Amhara Regional State.

Potash

- ✓ The potash reserve in the Danakil (Dallol Depression) of the Afar region is believed to be significant

GYPSUM AND ANHYDRITE

- ✓ A gypsum may be produced for domestic consumption in Ethiopia, mainly for the cement industry,
- ✓ but very large deposits are known to occur in sedimentary formations of the Red Sea coastal area, Danakil Depression, Ogaden, Shewa, Gojjam, Tigray, and Hararghe.
- ✓ Total reserves are probably enormous b/c the thickness of the gypsum deposits is many hundreds of meters and the formation are known to extend laterally for hundreds of kilometers.

CLAY

- ✓ Ethiopia is endowed with industrial clay material.
- ✓ Alluvial clay deposits for bricks and tile, pottery and pipe industry occur in Adola, Abay gorge, and the Rift Valley lakes region.
- ✓ Ceramic clay for the production of glasses, plates, bricks is found at Ambo and Adola.
- ✓ Tabor ceramic industry in Hawassa gets most of its raw materials from local sources.

MARBLE

- ✓ Crystalline limestone is widespread in the basement rocks of Ethiopia.
- ✓ Marble has been quarried in such localities as west of Mekelle and south of Adwa in Tigray.
- ✓ In the east in Galetti, Soka, Ramis, Rochelle, Kumi and other valleys of Chercher Mountain in West Hararghe.
- ✓ In the northwestern also in areas built of Precambrian schist in Gonder, and the Dabus River and other neighboring river basins in Benishangul-Gumuz and Gojjam.

CONSTRUCTION STONES

- ✓ Basalt, granite, limestone and sandstone are important building stones.
- ✓ For the surfacing of roads and compaction, basalt, scoria and other volcanic rocks are extensively used.
- ✓ Mesozoic limestone is an important raw material for cement and chalk production.
- ✓ The earlier cement works at Dire Dawa and the recent ones at Muger Valley, Abay gorge (Dejen), Tigray (Messebo) are using similar raw materials from these rock formations.

2.5.2. MINERAL POTENTIAL SITES OF ETHIOPIA

- ✓ According to the Ethiopian geological survey, the formations that host most mineral potentials of Ethiopia includes 3 major greenstone belts and other formations (Fig. 2.3). these are:

1. **The Western and South-western-greenstone belt:**

- ✓ They contain d/t minerals like gold occurrences (Dul,Tulu-Kape,Oda-Godere Akobo, Baruda,Bekuji-Motish and Kalaj);
- ✓ Yubdo Platinum,
- ✓ Base metals of AzaliAkendeyu,
- ✓ Abetselo and Kata;Fakushu Molybdenite and
- ✓ the iron deposits of Bikilal, Chago, Gordana and Korre, Benshagul-Gumuz-Marble, Akobo and Asosa placer gold deposits and etc.

2.5.2. MINERAL POTENTIAL SITES

2. The Southern greenstone belt:

- ✓ It is known as the Adola belt,
- ✓ which comprises the primary gold deposits and occurrences of Legadembi, Sakaro, Wellena, Kumudu, Megado-Serdo, Dawa Digati, Moyale and Ababa River;
- ✓ the columbo-tantalite of kenticha and Meleka, and the Adola nickel deposit and other industrial minerals.

2.5.2 MINERAL POTENTIAL SITE....

3. The Northern greenstone belt(Tigray):

- ✓ This belt comprises of the primary gold occurrences of Terakemti, Adi-Zeresenay, and Nirague.
- ✓ The base metals of Terer, Tsehafiembra and other parts of Tigray, Placer gold occurrences of Tigray.

